

Use Cases in ML

- **Training objectives:**
 - KL in variational inference (ELBO).
 - JS in GANs.
 - χ^2 divergences in robust estimation.
- **Two-sample testing / goodness-of-fit:**
 - Use $D_f(\alpha||\beta)$ as test statistic.
- **Regularization:** penalizing divergence from prior or reference distribution.
 - Eg TRPO, PPO in RL
- **Unsupervised learning:**
 - Hellinger distance in clustering & classification.
 - TV divergence in image reconstruction / denoising.

Part II: Integral Probability Metrics